Miryala Sriman

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SUMMARY

Software Engineer Apprentice at DBS Bank and B.Tech CS student at Vasavi College of Engineering (CGPA 8.03, graduating June 2027). Strong in data structures and algorithms (LeetCode Knight, **top 3% globally**; Codeforces Specialist). Built a real-time distributed platform (**40%** lower connection overhead, **sub-50ms** sync, **60%** faster deployments) and a PyTorch model that runs **350x faster**, using **Python** and **Java**.

EXPERIENCE

Software Engineer Apprentice — DBS Bank Technology Jun 2026 – Present

- Completing labs on **application security** and common cyber attacks in the DBS apprenticeship program
- Learning enterprise tech stacks like **Spring Boot** and **DevOps** through structured training

PROJECTS

Real-Time Collaborative Coding Platform *FastAPI, Redis, WebSockets, React.js, Docker, GitHub Actions*

- Reduced connection overhead by **40%** in load tests by adding a Redis pub/sub layer to sync user sessions
- Kept sync latency under **50ms** (WebSocket round-trip time) by designing and debugging real-time editing
- Made deployments **60% faster** by using Docker containers and automating CI/CD with GitHub Actions
- Ranked **Top 10 out of 100+ teams** at GenAI Hackathon 2025 by building a live multi-user coding product

Constraint-Based Scheduling Engine *Python, CSP, Graph Coloring, AC-3*

- Modeled university timetabling as a CSP with graph coloring across **50+ hard constraints**
- Used backtracking search to find valid timetables for the CSP
- Applied AC-3 arc consistency to cut down the search space, checked with runtime benchmarks
- Built an automated scheduler with **99.7% conflict-free** timetables, **60% faster** than brute force

ResU-Net Surrogate for Topology Optimization *Python, PyTorch, CUDA, React.js, Google Colab*

- Reached a **Dice score of 0.89** against ground-truth simulations by training and evaluating the model
- Improved training speed **10x** (samples per second) by using CUDA Automatic Mixed Precision
- Owned the **full ML pipeline** (preprocessing, training, evaluation, inference) from raw data to results
- Made topology optimization **350x faster** than the old optimizer using a PyTorch encoder-decoder model

EDUCATION

B.Tech, Computer Science and Engineering — Vasavi College of Engineering Sep 2023 – Jun 2027

- GPA: 8.03
- Coursework: DSA, Algorithm Analysis, Operating Systems, Distributed Systems, DBMS, Computer Networks

Intermediate (MPC) — Narayana Junior College, Hyderabad | 98.1% Aug 2021 – Jul 2023

SKILLS

Languages: Python, Java, C++, C, JavaScript, SQL
Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, Distributed Systems, Computer Networks
Systems: Redis Pub/Sub, WebSockets, Multi-threading, Load Testing, Debugging
Backend & Web: FastAPI, Spring Boot, React.js, REST APIs, MySQL, MongoDB
AI/ML: PyTorch, CUDA, Deep Learning, Generative AI (OCI certified)
Tools: Docker, GitHub Actions (CI/CD), Git

ACHIEVEMENTS

- **LeetCode Knight** — max rating **1989, top 3% globally** (leetcode.com/u/Srimanmiryala)
- **Codeforces Specialist** — max rating **1503** (codeforces.com/profile/Srimanmiryala)
- Ranked **936 of 20,000+** in LeetCode Biweekly Contest 177 and **1,331 of 25,000+** in Weekly Contest 501
- TS EAMCET 2023 state rank **2,245 of 224,000+** candidates

CERTIFICATIONS

- Oracle Cloud Infrastructure Generative AI Professional — Oracle, 2025
- Python Essentials 1 & 2 — Cisco Networking Academy, 2025